Hugo Marquerie 27/02/2025

Desigualdad de Hölder

Teorema 1 (Desigualdad de Hölder). *Sea* $p \in (1, \infty)$ *y* $f \in \mathcal{L}^p(\mu), g \in \mathcal{L}^q(\mu)$

$$\implies \|f \cdot g\|_1 \leqslant \|f\|_p \cdot \|g\|_q$$

donde q *es el exponente conjugado de* p.

***Demostración*:** Si $\|f\|_p = 0$ o $\|g\|_q = 0$, la desigualdad es trivial, por tanto, suponemos que $\|f\|_p \neq 0$ y $\|g\|_q \neq 0$ y definimos $\widehat{f} := f/\|f\|_p \wedge \widehat{g} := g/\|g\|_q$.

$$\implies \|f \cdot g\|_1 = \int_\Omega |f(x) \cdot g(x)| \, \mathrm{d}\mu(x) = \|f\|_p \cdot \|g\|_q \int_\Omega \left|\widehat{f}(x) \cdot \widehat{g}(x)\right| \mathrm{d}\mu(x).$$

Por la desigualdad de Young, $\forall x \in \Omega : \left|\widehat{f}(x) \cdot \widehat{g}(x)\right| \leqslant \frac{\left|\widehat{f}(x)\right|^p}{p} + \frac{|\widehat{g}(x)|^q}{q}$.

$$\implies \|f \cdot g\|_1 \leqslant \|f\|_p \cdot \|g\|_q \left(\frac{1}{p} \int_\Omega \left|\widehat{f}(x)\right|^p \mathrm{d}\mu(x) + \frac{1}{q} \int_\Omega |\widehat{g}(x)|^q \, \mathrm{d}\mu(x)\right) =$$
$$\leqslant \|f\|_p \cdot \|g\|_q \left(\frac{1}{p} + \frac{1}{q}\right) = \|f\|_p \cdot \|g\|_q .$$

■

Referenciado en

- `teo-aproximacion-identidad-convolucion`
- `prop-inclusion-lp-general`
- `prop-convolucion-exp-conjugados`
- `desigualdad-minkowski`
- `prop-inclusion-lp-esp-finito`

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